

1. FCFS Scheduling algorithm

```
#include <stdio.h>
int main()
{
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i=0;i<n;i++){
        printf("P%d: ", i+1);
        scanf("%d", &bt[i]);
    }

    wt[0]=0;
    for(i=1;i<n;i++){
        wt[i]=wt[i-1]+bt[i-1];
    }

    for(i=0;i<n;i++){
        tat[i]=wt[i]+bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++){
        printf("P%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);
    }
    return 0;
}
```

Output:

```
shashank@LAPTOP-24KPR0R8:~$ gcc FCFS.c -o FCFS
shashank@LAPTOP-24KPR0R8:~$ ./FCFS
Enter number of processes: 2
Enter burst time for each process:
P1: 10
P2: 20

Process BT      WT      TAT
P1      10      0       10
P2      20      10      30
```

2. SJF_NONpreemptive

```
#include <stdio.h>
#include <stdio.h>
int main(){
int n, i, j;
int bt[20], wt[20], tat[20];
int temp;
printf("Enter number of processes: ");
scanf("%d", &n);

printf("Enter burst time:\n");
for (i = 0; i < n; i++) {
printf("P%d: ", i + 1);
scanf("%d", &bt[i]);
}

for (i = 0; i < n - 1; i++) {
for (j = i + 1; j < n; j++) {
if (bt[i] > bt[j]) {
temp = bt[i];
bt[i] = bt[j];
bt[j] = temp;
}
}
}
wt[0] = 0;

for (i = 1; i < n; i++) {
wt[i] = wt[i - 1] + bt[i - 1];
}

for (i = 0; i < n; i++) {
tat[i] = wt[i] + bt[i];
}
printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);
}
return 0;
}
```

Output:

```

shashank@LAPTOP-24KPROR8:~$ gcc SJFnp.c -o SJFnp
shashank@LAPTOP-24KPROR8:~$ ./SJFnp
Enter number of processes: 2
Enter burst time:
P1: 15
P2: 20

Process  BT      WT      TAT
P1       15      0       15
P2       20     15      35

```

3. Priority

```

#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter burst time and priority:\n");
    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");
}

```

```

printf("\nProcess\tBT\tPriority\tWT\tTAT\n");
for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t\t%d\t%d\n",
           i + 1, bt[i], priority[i], wt[i], tat[i]);
}
return 0;
}

```

Output:

```

shashank@LAPTOP-24KPR0R8:~$ gcc Priority.c -o Priority
shashank@LAPTOP-24KPR0R8:~$ ./Priority
Enter number of processes: 2
Enter burst time and priority:
P1 Burst Time: 5
P1 Priority: 10
P2 Burst Time: 10
P2 Priority: 15

Process BT      Priority      WT      TAT
P1      5        10          0        5
P2     10        15          5       15

```

4. ROUND ROBIN

```
#include <stdio.h>
int main() {
int n, i;
int bt[20], rem_bt[20];
int wt[20], tat[20];
int quantum;
int time = 0;
int done;

printf("Enter number of processes: ");
scanf("%d", &n);

printf("Enter burst time:\n");
for (i = 0; i < n; i++) {
    printf("P%d: ", i + 1);
    scanf("%d", &bt[i]);

    rem_bt[i] = bt[i];
    wt[i] = 0;
}

printf("Enter time quantum: ");
scanf("%d", &quantum);

while (1) {
    done = 1;

    for (i = 0; i < n; i++) {
        if (rem_bt[i] > 0) {
            done = 0;

            if (rem_bt[i] > quantum) {
                time = time + quantum;
                rem_bt[i] = rem_bt[i] - quantum;
            }
        }
    }
}
```

```

        else {
            time = time + rem_bt[i];

            wt[i] = time - bt[i];

            rem_bt[i] = 0;
        }
    }

    if (done == 1)
        break;
}

for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
           i + 1, bt[i], wt[i], tat[i]);
}
return 0;
}

```

Output:p

```

shashank@LAPTOP-24KPR0R8:~$ gcc Roundrobin.c -o Roundrobin
shashank@LAPTOP-24KPR0R8:~$ ./Roundrobin
Enter number of processes: 2
Enter burst time:
P1: 12
P2: 14
Enter time quantum: 6

Process BT      WT      TAT
P1      12      6      18
P2      14     12     26

```